Tagging efficiency vs compute across the activation-quantization axis

Era-2 HLS4ML LHC Jet 5-class · held-out split n = 260,000 · trained refs = seed s1 · the two EBOPs conventions are separate columns, never mixed

Model	Trained AUC ROC-test macro-OvR	HGQ2 rebuild AUC binary-pinned, static act quant	Δ (rebuild − trained)	EBOPs (HGQ2-native) accumulator in	EBOPs (analytic) no accumulator
FP32 (vanilla)	0.8765	—	—	—	64.95 G
W8A8	0.8642	—	—	—	4.06 G
W1A8 (binary, 8-bit act)	0.8551	**0.8493**	-0.0058	650.4 M	530.4 M
W1A6 (binary, 6-bit act)	0.8394	**0.8222**	-0.0173	501.2 M	392.9 M
W1A4 (binary, 4-bit act)	0.7346	**0.7115**	-0.0231	352.2 M	258.6 M

HGQ2-native EBOPs exist only for the three rebuilt binary models (no HGQ2 build of FP32/W8A8). Within the analytic convention only: W8A8 / W1A8 = 7.65×.

Large model (D256/L8), per-shape/per-probe measurements — provisional; deployable single-model table awaits r6-small

Sources: roc-results/r5/*.npz · results/hgq2/runs/*/scores.npz · results/hgq2/runs/*/ebops.json · results/ebops.md (re-derived) — poster/VERIFICATION.md §1–§2